Week 13 - Objectives Normalization and PF distance

Let $\mathcal{J} = \{1, 2, \ldots, n\}$ be the batch of jobs arriving over the time horizon T.
Let $\mathcal{I} = \{\text{CPU}, \text{RAM}\}$ be the set of physical resources.
$x_j \in [0, 1]$: The fractional acceptance variable for job j.

Parameters per Job j:

- $q_j \in \{1, 2, 3, 4, 5\}$: Ordinal priority tier (mapped to prevent explosion).
- v_j: Customer continuous valuation.
- $\Phi(L_j)$: Myerson Virtual Value evaluated at the lower bound of the DP grid bin.
- $C_{elec,j}$: Expected electricity cost ($w_j \cdot D_j \cdot c_{elec}$).
- $C_{carb,j}$: Expected carbon cost ($w_j \cdot D_j \cdot E_t \cdot SCC$).
- $A_{i,j}$: Instantaneous demand for resource i.
- D_j: Total job duration.

The Feasible Decision Space ($\mathcal{X}$):

The optimizer defines the feasible space using **Fluid Volume Constraints** rather than instantaneous scheduling constraints to provide a tractable upper bound (OPT_{LP}) for the DLENT regret analysis:

$$\mathcal{X} = \left\{ \mathbf{x} \in [0, 1]^n \quad \sum_{j=1}^{n} x_j A_{i,j} D_j \leq c_i \cdot T \quad \forall i \in \mathcal{I} \right\}$$

Raw objective functions

Before normalization, the class defines three distinct, independent objective vectors $\mathbf{c}_{sat}$, $\mathbf{c}_{prof}$, and $\mathbf{c}_{carb}$:

1. **Customer Satisfaction:** $f_{sat}(\mathbf{x}) = \sum_{j=1}^{n} x_j (q_j v_j)$
2. **Provider Profit:** $f_{prof}(\mathbf{x}) = \sum_{j=1}^{n} x_j (\Phi(L_j) - C_{elec,j})$
3. **Carbon Cost (Minimization):** $f_{carb}(\mathbf{x}) = -\sum_{j=1}^{n} x_j C_{carb,j}$

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1. Normalization

Ideal and Nadir computations
Because $f_{sat} \sim 10^7$ while $f_{carb} \sim 10^3$, any unscaled aggregation will be dominated by satisfaction. The `_compute_ideal_and_nadir()` method find the theoretical limits of the system:

1. **Ideal Satisfaction:** $z^*_{sat} = \max_{\mathbf{x} \in \mathcal{X}} f_{sat}(\mathbf{x})$
2. **Ideal Profit:** $z^*_{prof} = \max_{\mathbf{x} \in \mathcal{X}} f_{prof}(\mathbf{x})$
3. **Nadir Carbon Cost:** $z^{nadir}_{carb} = \max_{\mathbf{x}^* \in \mathcal{P}^*} f_{carb}(\mathbf{x}^*)$
4. **The Ideal Point (z^*):** The absolute *best* possible outcome for an objective if we optimized for it alone, completely ignoring the others.
5. **The Nadir Point (z^{nadir}):** The absolute *worst* possible outcome for an objective while still operating within the feasible space of the Pareto frontier.

Carbon Cost (f_{carb}) is a penalty. We want to **minimize** it.

- The Ideal scenario is easy: $\$0$ carbon cost (achieved by rejecting every single job)

- But we cannot divide by 0 to normalize it. We need a ceiling $\rightarrow$ **Nadir Carbon Cost**.

In our context, the Nadir Carbon Cost (z_{carb}^{nadir}) is the **maximum possible environmental the cluster could inflict over the month of May, given its physical capacity constraints.**

Mathematically, we find it by telling the LP solver to actively **maximize** the carbon cost, filling up CPU and RAM capacity ($c_i \cdot T$) with the dirtiest, most power-hungry jobs available in the dataset, completely ignoring whether those jobs are profitable or high-priority.

$$z_{carb}^{nadir} = \max_{\mathbf{x} \in \mathcal{X}} \sum_{j=1}^{n} x_j \cdot C_{carbon,j}$$

Once we know the Nadir and the Ideal (zero), we can mathematically define the normalized Chebyshev distance.

To normalize this into a dimensionless $[0, 1]$ scale so it can compete fairly with Profit and Satisfaction, we divide it by the total distance between the Nadir and the Ideal points:

$$\text{Normalized Carbon Distance} = \frac{f_{carb}(\mathbf{x}) - z_{carb}^{*}}{z_{carb}^{nadir} - z_{carb}^{*}}$$

Normalized scalarization methods

We map all objectives from their raw monetary space $()intoastrictlydimensionless\$[0, 1]$ space.

A. Normalized linear scalarization

Instead of maximizing the raw weighted sum, the algorithm maximizes the weighted sum of the *percentage of potential achieved*.

$$\max_{\mathbf{x} \in \mathcal{X}} \quad \lambda_1 \left(\frac{f_{sat}(\mathbf{x})}{z_{sat}^{*}} \right) + \lambda_2 \left(\frac{f_{prof}(\mathbf{x})}{z_{prof}^{*}} \right) - \lambda_3 \left(\frac{f_{carb}(\mathbf{x})}{z_{carb}^{nadir}} \right)$$

This guarantees that if $\lambda_1 = \lambda_2 = \lambda_3$, an incremental 1% gain in theoretical satisfaction is treated exactly the same mathematically as a 1% gain in theoretical profit or a 1% reduction in maximum carbon, completely eliminating the $30\times$ magnitude bias.

B. Normalized chebyshev scalarization (L_∞)

Chebyshev scalarization seeks to minimize the maximum weighted distance from the chosen solution to the Ideal point.

Objective:

$$\min_{\mathbf{x}, t} \quad t$$

Subject to the Distance Constraints:

1. **Satisfaction Distance:** $\lambda_1 \left(1 - \frac{f_{sat}(\mathbf{x})}{z_{sat}^{*}}\right) \leq t$
2. **Profit Distance:** $\lambda_2 \left(1 - \frac{f_{prof}(\mathbf{x})}{z_{prof}^{*}}\right) \leq t$
3. **Carbon Distance:** $\lambda_3 \left(\frac{f_{carb}(\mathbf{x}) - 0}{z_{carb}^{nadir}}\right) \leq t$
4. **Fluid Capacity:** $\mathbf{x} \in \mathcal{X}$

Pareto front distance

We have two Pareto fronts in $\mathbb{R}^3$:

$$\mathcal{A} = \text{online mechanism's front}, \qquad \mathcal{Z}^* = \text{the Fluid LP upper-bound front}$$

The question *"how far apart are they?"* is hard. Unlike comparing two scalar values, comparing two *sets of points in objective space* requires answers to three simultaneous sub-questions:

Sub-question	Mathematical character	Indicator that answers it
How much of the objective space does $\mathcal{A}$ dominate?	Volumetric	**Hypervolume (HV)**
How close is every LP-front point to $\mathcal{A}$?	Average point-to-set distance	**IGD⁺**
What is the worst-case dominance gap?	Max-min distance	**ε-indicator**
What fraction of $\mathcal{Z}^*$ does $\mathcal{A}$ cover?	Set-ratio	**C-metric**
Is $\mathcal{A}$ spread uniformly?	Diversity	**Spacing / Spread**

No single number answers all five simultaneously.

Formal Setup and Pareto-Compliance

$$\text{Decision space: } \mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^n$$
$$\text{Objective space: } \mathbf{F}(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x})) \in \mathbb{R}^m$$

An **approximation set** $\mathcal{A} \subset \mathbb{R}^m$ is a finite, mutually non-dominated set of objective vectors. This general framework works in **minimization** throughout (Later I'll re-casts everything for maximization problem).

Set dominance

$$\mathcal{A} \trianglelefteq \mathcal{B} \quad \iff \quad \forall \mathbf{b} \in \mathcal{B},\ \exists \mathbf{a} \in \mathcal{A} : \mathbf{a} \preceq \mathbf{b} \qquad (\mathcal{A} \text{ weakly dominates } \mathcal{B})$$
$$\mathcal{A} \triangleleft \mathcal{B} \quad \iff \quad \mathcal{A} \trianglelefteq \mathcal{B} \text{ and } \mathcal{A} \neq \mathcal{B} \qquad (\mathcal{A} \text{ strictly dominates } \mathcal{B})$$

The compliance hierarchy

Before we trust an indicator, we have to run it through a mathematical lie detector test called **Pareto Compliance**.
If our online mechanism ($\mathcal{A}$) is strictly better than a baseline algorithm ($\mathcal{B}$), meaning every single solution in $\mathcal{B}$ is beaten by at least one solution in $\mathcal{A}$, the indicator **must** give $\mathcal{A}$ a better score.

Definition (Zitzler, Thiele, Laumanns, Fonseca, Grunert da Fonseca — IEEE TEVC 2003): A unary indicator $I : 2^{\mathbb{R}^m} \to \mathbb{R}$ is:

$$\textbf{Strictly Pareto-compliant} \iff \mathcal{A} \triangleleft \mathcal{B} \implies I(\mathcal{A}) \succ I(\mathcal{B})$$

If $\mathcal{A}$ strictly dominates $\mathcal{B}$, the score for $\mathcal{A}$ will always be strictly better.

$$\textbf{Weakly Pareto-compliant} \iff \mathcal{A} \trianglelefteq \mathcal{B} \implies I(\mathcal{A}) \succeq I(\mathcal{B})$$

The metric *usually* tells the truth, but sometimes it might say $\mathcal{A}$ and $\mathcal{B}$ are tied, even if $\mathcal{A}$ is slightly better.

$$\textbf{Pareto non-compliant} \iff \exists\, \mathcal{A} \triangleleft \mathcal{B} \text{ s.t. } I(\mathcal{A}) \preceq I(\mathcal{B})$$

The metric is fundamentally flawed. It might occasionally tell you that $\mathcal{B}$ is better than $\mathcal{A}$, even when $\mathcal{A}$ mathematically dominates it.

where $\succ$ means "strictly better" (greater for HV, smaller for distances). Every indicator below will be classified against it, with explicit counter-examples where it fails.

- **Hypervolume (HV) is the only strictly Pareto-compliant unary indicator that exists.** That is why it is usually the primary metric in any multi-objective paper.

Because we are dealing with a **minimization** problem, "better" means pushing the points as close to zero (the bottom-left) as possible.

Indicator 1: The Hypervolume Indicator

The general idea of this inidicator is:

- **Reference point ($\mathbf{r}$):** we place a dot in the top-right corner of the graph. This represents a terrible baseline score.
- Draw a box extending from every point in the Pareto front up and to the right, until it hits the reference point.
- The total area (or volume in 3D) covered by those overlapping boxes is the Hypervolume. The bigger the shadow, the better our front is.

Definition

For an approximation set $\mathcal{A} = \{\mathbf{a}^{(1)}, \ldots, \mathbf{a}^{(n)}\} \subset \mathbb{R}^m$ and a **reference point** $\mathbf{r} \in \mathbb{R}^m$ satisfying $\mathbf{a}^{(i)} \preceq \mathbf{r}$ for all i, the Hypervolume Indicator is:

$$\mathrm{HV}(\mathcal{A}, \mathbf{r}) = \lambda_m \left(\bigcup_{i=1}^{n} \left[\mathbf{a}^{(i)}, \mathbf{r} \right] \right)$$

where λ_m is the m-dimensional Lebesgue measure and $[\mathbf{a}, \mathbf{r}] = \{\mathbf{y} \in \mathbb{R}^m : \mathbf{a} \preceq \mathbf{y} \preceq \mathbf{r}\}$ is a hyper-rectangle. In plain language: **HV is the volume of the region jointly dominated by all points in $\mathcal{A}$ and bounded above by $\mathbf{r}$.**

The **exclusive hypervolume contribution** (HVC) of a single point $\mathbf{a}^{(k)} \in \mathcal{A}$ is:

$$\mathrm{HVC}(\mathbf{a}^{(k)}, \mathcal{A}, \mathbf{r}) = \mathrm{HV}(\mathcal{A}, \mathbf{r}) - \mathrm{HV}(\mathcal{A} \setminus \{\mathbf{a}^{(k)}\}, \mathbf{r})$$

This is the volume uniquely attributable to $\mathbf{a}^{(k)}$, the region it dominates that no other point in $\mathcal{A}$ dominates.

If we delete one point from the front, the total shadow will shrink. The exact amount of shadow that disappears is that point's **HVC**. It tells us how uniquely valuable that specific point is. If a point is sitting right next to another point, its HVC is tiny (it is redundant). If it is sitting alone in a deep gap, its HVC is massive.

Theorem (Zitzler et al. 2003): The Hypervolume Indicator is **strictly Pareto-compliant**.

Reference-point selection

The reference point $\mathbf{r}$ is the most consequential parameter. Three theoretically grounded choices:

Choice 1 — Nadir point plus offset (standard):

$$\mathbf{r} = \mathbf{z}^{\mathrm{nadir}} + \epsilon \cdot \mathbf{1}, \qquad \epsilon = 0.1 \cdot \|\mathbf{z}^{\mathrm{ideal}} - \mathbf{z}^{\mathrm{nadir}}\|_\infty$$

Choice 2 — Fixed normalized offset (Ishibuchi et al. 2018, *Evol. Comput.* 26(3):411):

$$\mathbf{r} = \mathbf{z}^{\mathrm{nadir}} + \frac{1}{H-1}(\mathbf{z}^{\mathrm{nadir}} - \mathbf{z}^{\mathrm{ideal}})$$

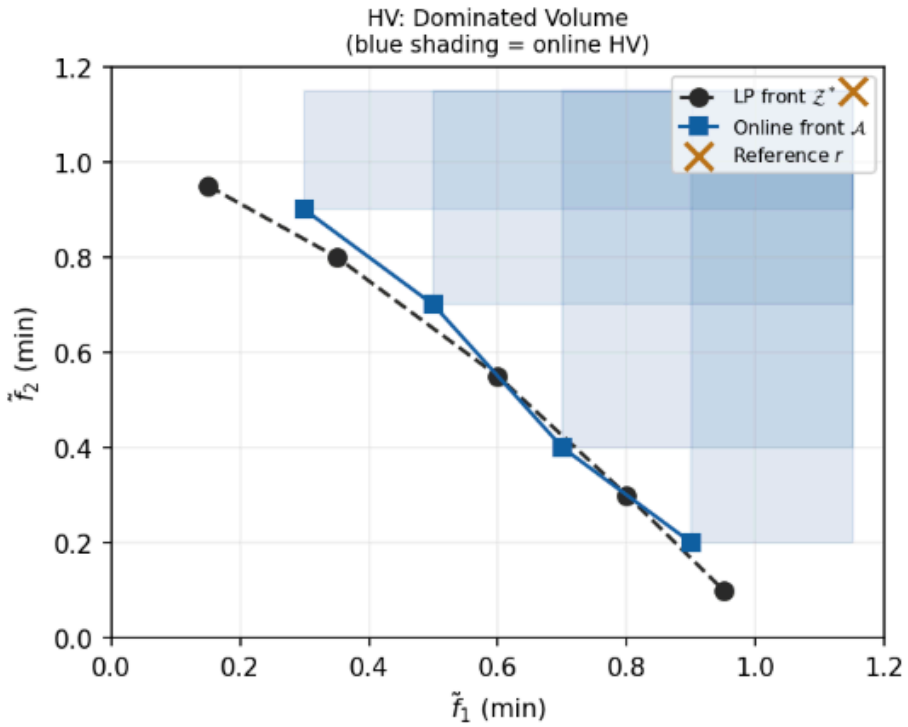
where H is the archive size. For archives of 30–100 and m=3 this gives $\approx z_i^{\mathrm{nadir}} + 0.033 - 0.1$ per dimension, matching the conventional choice $\mathbf{r} = (1.1, 1.1, 1.1)$

Choice 3 — Worst-observed across all runs:

$$r_i = \max_{\text{run } k} \max_{a \in \mathcal{A}^{(k)}} a_i + \epsilon$$

This guarantees the same $\mathbf{r}$ is valid across all algorithm variants, enabling fair comparison.

We add that little ϵ offset because if we put the reference point exactly *on* the worst solution (the Nadir point), the box drawn from that solution will have a width or height of zero, making its volume zero. By adding an offset (like $+0.1$), we give every point a non-zero volume, allowing the math to work properly.



Indicator 2: Generational Distance (GD) and GD⁺

Basic idea: When we build an algorithm to find the Pareto front, we want to know: **"How physically close did my points get to the true, perfect Pareto front?"**

To calculate GD, we do the following:

1. Take a point the algorithm found (let's call it A).
2. Look at the true, perfect Pareto front ($\mathcal{Z}^*$) and find the single closest perfect point to A.
3. Measure the physical distance between them using a standard straight line (Euclidean distance).
4. Average that distance across all the points the algorithm found.

Lower GD is better (zero means the points are exactly sitting on the perfect front).

Introduced by Van Veldhuizen (AFIT PhD dissertation, 1999) and formalized in Van Veldhuizen & Lamont (CEC 2000):

$$\mathrm{GD}_p(\mathcal{A}, \mathcal{Z}^*) = \frac{1}{|\mathcal{A}|} \left(\sum_{i=1}^{|\mathcal{A}|} \min_{\mathbf{z} \in \mathcal{Z}^*} \|\mathbf{a}^{(i)} - \mathbf{z}\|_2^p \right)^{1/p}$$

The standard choice is p=1 (mean distance) or p=2 (RMS distance). **Lower is better.** GD measures *convergence only*: it rewards points in $\mathcal{A}$ close to $\mathcal{Z}^*$ but says nothing about whether $\mathcal{A}$ covers $\mathcal{Z}^*$ uniformly.

GD is Pareto non-compliant: counter-example

Let $\mathcal{Z}^* = \{(1,0),(0,10)\}$, $\mathcal{A} = \{(2,5)\}$, $\mathcal{B} = \{(3,9)\}$

- $\mathbf{a} = (2,5)$ strictly dominates $\mathbf{b} = (3,9)$, so $\mathcal{A} \lhd \mathcal{B}$.
- $\mathrm{GD}(\mathcal{A}, \mathcal{Z}^*) = \min(\sqrt{(2-1)^2+5^2}, \sqrt{4+25}) = \sqrt{26} \approx 5.10$
- $\mathrm{GD}(\mathcal{B}, \mathcal{Z}^*) = \min(\sqrt{4+81}, \sqrt{9+1}) = \sqrt{10} \approx 3.16$

GD ranks $\mathcal{B}$ **better** than $\mathcal{A}$ despite $\mathcal{A}$ strictly dominating $\mathcal{B}$. Any methodology using GD as a primary ranking criterion can publish inverted conclusions. □

GD⁺: the compliance-motivated fix

Ishibuchi, Masuda, Tanigaki, Nojima (EMO 2015, LNCS 9019:110–125, DOI 10.1007/978-3-319-15892-1*8) replace the Euclidean distance with the **modified distance**:*
*$$d^+(\mathbf{z}, \mathbf{a})* = \sqrt{\sum{k=1}^{m} \left(\max{a_k - z_k,\ 0}\right)^2}$$

Geometrically: $d^+(\mathbf{z}, \mathbf{a})$ is zero if $\mathbf{a} \preceq \mathbf{z}$ (a dominates z — ideal situation), and equals the Euclidean distance only in directions where $\mathbf{a}$ is *worse* than $\mathbf{z}$. It projects the gap onto the non-dominated orthant.

$$\mathrm{GD}^+(\mathcal{A}, \mathcal{Z}^*) = \frac{1}{|\mathcal{A}|} \sum_{i=1}^{|\mathcal{A}|} \min_{\mathbf{z} \in \mathcal{Z}^*} d^+(\mathbf{z},\ \mathbf{a}^{(i)})$$

GD⁺ resolves the counter-example above: and GD⁺ correctly ranks $\mathcal{A}$ better. However GD⁺ still suffers from **asymmetry**, it measures how well $\mathcal{A}$ converges toward $\mathcal{Z}^*$ but not how well it covers $\mathcal{Z}^*$.

Indicator 3: Inverted Generational Distance (IGD and IGD⁺)

GD measures distances *from $\mathcal{A}$ to $\mathcal{Z}^*$* (convergence). IGD inverts the direction, measuring distances *from $\mathcal{Z}^*$ to $\mathcal{A}$* — which simultaneously captures convergence **and diversity**:

$$\mathrm{IGD}_p(\mathcal{A}, \mathcal{Z}^*) = \frac{1}{|\mathcal{Z}^*|} \left(\sum_{j=1}^{|\mathcal{Z}^*|} \min_{\mathbf{a} \in \mathcal{A}} \|\mathbf{z}^{(j)} - \mathbf{a}\|_2^p \right)^{1/p}$$

First proposed by Bosman & Thierens (IEEE TEVC 2003, DOI 10.1109/TEVC.2003.810761) and named IGD by Coello Coello & Reyes-Sierra (MICAI 2004, DOI 10.1007/978-3-540-24694-7_71).

Standard GD has a massive blind spot: **It completely ignores diversity.** If the perfect Pareto front ($\mathcal{Z}^*$) is a massive curve spanning from 0 to 100, and your algorithm only finds three points clustered tightly together at the very top edge, GD might say your algorithm is "perfect" because those three points happen to sit exactly on the line. It doesn't care that you completely missed 99% of the curve.
$\rightarrow$ If $\mathcal{A}$ is concentrated in one region of $\mathcal{Z}^*$, many reference points $\mathbf{z}^{(j)}$ in uncovered regions will have large nearest-neighbor distances to $\mathcal{A}$, inflating IGD even if $\mathcal{A}$ has excellent convergence in its covered region.

IGD flips the script:

1. It looks at every single perfect point in the true front ($\mathcal{Z}^*$).
2. From *each* perfect point, it draws a line to the nearest point the algorithm found ($\mathcal{A}$).
3. It averages all those distances.

So, if the algorithm's points are clustered in one tiny corner, the perfect points on the far opposite end of the curve will have to draw massively long lines to reach them. These long distances will ruin IGD score. To get a good IGD score (close to zero), the algorithm's points must be **both close to the perfect line AND spread evenly across it.**

IGD is still Pareto non-compliant: counter-example

Ishibuchi et al. 2015, Example 4. Let $\mathcal{Z}^* = \{(0,10),(1,6),(2,2),(6,1),(10,0)\}$. Consider:

- $\mathcal{A} = \{(2,4),\ (3,3),\ (4,2)\}$
- $\mathcal{B} = \{(2,8),\ (4,4),\ (8,2)\}$

Every point of $\mathcal{B}$ is dominated by a corresponding point of $\mathcal{A}$ ($(2,4) \prec (2,8)$, $(3,3) \prec (4,4)$, $(4,2) \prec (8,2)$), so $\mathcal{A} \triangleleft \mathcal{B}$.

However the distances from each $\mathbf{z}^{(j)}$ to the *nearest point in* $\mathcal{B}$ happen to be smaller in Euclidean norm because $\mathcal{B}$'s points lie closer to the reference-set extremes despite dominating worse. Computing IGD ($p = 1$) gives $\mathrm{IGD}(\mathcal{A}) > \mathrm{IGD}(\mathcal{B})$ — the dominated set $\mathcal{B}$ scores better. The root cause: Euclidean distance is not aligned with the dominance direction.

IGD⁺: the weak-compliance fix

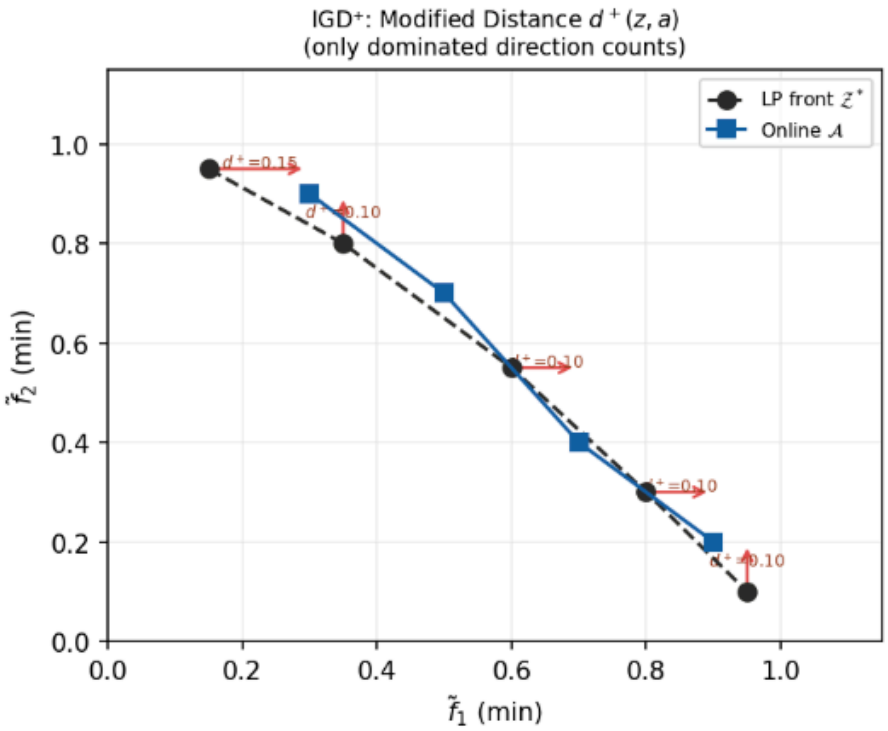
Apply d^+ symmetrically — from $\mathcal{Z}^*$ to $\mathcal{A}$:

$$\boxed{\mathrm{IGD}^+(\mathcal{A}, \mathcal{Z}^*) = \frac{1}{|\mathcal{Z}^*|} \sum_{j=1}^{|\mathcal{Z}^*|} \min_{\mathbf{a} \in \mathcal{A}} d^+(\mathbf{z}^{(j)}, \mathbf{a})}$$

where $d^+(\mathbf{z}, \mathbf{a}) = \sqrt{\sum_k \max\{a_k - z_k, 0\}^2}$.

Theorem (Ishibuchi et al. 2015): IGD⁺ is weakly Pareto-compliant.

IGD⁺ is only *weakly* compliant — when $\mathcal{A} \triangleleft \mathcal{B}$ strictly, IGD⁺ may still give $\mathrm{IGD}^+(\mathcal{A}) = \mathrm{IGD}^+(\mathcal{B})$ if the strict improvement occurs at a point not covered by $\mathcal{Z}^*$. This is an acceptable limitation when $\mathcal{Z}^*$ is dense.



Indicator 4: The Additive ε-Indicator

If Hypervolume measures **total mass** and IGD measures **average distance**, the Additive ε-Indicator measures the **worst-case scenario**.
Instead of asking "how close did we get on average?", it asks: **"What is the single biggest failure of my algorithm, and exactly how much cheating would it take to fix it?"**

Zitzler, Thiele, Laumanns, Fonseca, Grunert da Fonseca (IEEE TEVC 2003) define a family of binary indicators. The additive unary ε-indicator is:

$$I_{\varepsilon^+}(\mathcal{A}, \mathcal{Z}^*) = \max_{\mathbf{z} \in \mathcal{Z}^*} \min_{\mathbf{a} \in \mathcal{A}} \max_{k=1,\ldots,m} (a_k - z_k)$$

1. `max(a_k - z_k)` **:** Take one point the algorithm found (A) and compare it to one perfect point (Z). Look at all dimensions (X, Y, Z). Which dimension has the biggest gap where A is worse than Z? This is the "Chebyshev distance", the worst individual coordinate difference between those two points.
2. `min_{a}` **:** Now, look at *all* the points the algorithm found ($\mathcal{A}$). Which one does the best job of covering Z? (i.e., which one has the smallest worst-case gap?).
3. `max_{z}` **:** Finally, look at *all* the perfect points in $\mathcal{Z}^*$. Which perfect point is the absolute hardest to cover? The gap for this specific point is the final ε-Indicator score.

Geometric interpretation: $I_{\varepsilon^+}(\mathcal{A}, \mathcal{Z}^*)$ is the smallest uniform shift ε such that, after translating every point of $\mathcal{A}$ by $-\varepsilon$ in every coordinate, $\mathcal{A} - \varepsilon\mathbf{1}$ weakly dominates $\mathcal{Z}^*$.
Imagine the algorithm's points ($\mathcal{A}$) and the perfect points ($\mathcal{Z}^*$) plotted on a graph. Because this is a minimization problem, the perfect points are closer to zero (bottom-left), and the algorithm's points are hovering somewhere above and to the right of them.

Now, imagine taking your entire set of points and physically sliding them diagonally down and to the left.

- The ε-Indicator is the **exact minimum distance you need to slide your front** until its "dominance shadow" (the area above and to the right of your points) completely swallows every single perfect point in $\mathcal{Z}^*$.
- If $\varepsilon = 0.5$, it means your worst-case gap is 0.5. You would have to artificially subtract 0.5 from *every single objective* of your algorithm before it could successfully dominate the true Pareto front.
- If $\varepsilon \leq 0$, it means you don't have to slide it at all. Your algorithm already perfectly dominates the reference se

Value	Meaning
$I_{\varepsilon^+} \leq 0$	$\mathcal{A}$ weakly dominates $\mathcal{Z}^*$ entirely
$I_{\varepsilon^+} = \delta > 0$	The worst gap: $\mathcal{A}$ must improve by δ on at least one objective to dominate the hardest reference point
$I_{\varepsilon^+} \gg 0$	$\mathcal{A}$ is far from the LP front in a worst-case sense

The **multiplicative form** (for positive coordinates) replaces $a_k - z_k$ with a_k / z_k:
$I_\varepsilon(\mathcal{A}, \mathcal{Z}^*) = \max_{\mathbf{z}} \min_{\mathbf{a}} \max_k (a_k / z_k)$. This measures the worst-case *ratio* gap, interpretable as a competitive ratio.

The binary ε-indicator is strictly Pareto-compliant: $\mathcal{A} \triangleleft \mathcal{B} \iff I_{\varepsilon^+}(\mathcal{A}, \mathcal{B}) \leq 0 \wedge I_{\varepsilon^+}(\mathcal{B}, \mathcal{A}) > 0$. The unary form $I_{\varepsilon^+}(\mathcal{A}, \mathcal{Z}^*)$ is weakly compliant. It guarantees that if Algorithm A is strictly better than Algorithm B, Algorithm A's score will be better (or tied).

Complexity
$O(|\mathcal{A}| \cdot |\mathcal{Z}^*| \cdot m)$ — no exponential term, exact, deterministic. For your $m = 3$ this is trivially fast.

DLENT environment

Our environment establishes a three-level hierarchy of Pareto fronts:

$$\underbrace{\mathcal{PF}^{\text{LB}}}_{\text{Random lower bound}} \leq \underbrace{\mathcal{PF}^{\text{Online}}}_{\text{your mechanism}} \leq \underbrace{\mathcal{PF}^{\text{UB}}}_{\text{Fluid LP upper bound}}$$

where $\leq$ means "weakly dominated by" in the set-dominance sense $\trianglelefteq$. This structure gives us a complete regret decomposition:

$$\underbrace{\mathrm{Gap}_{\mathrm{total}}(\mathbf{x}^{\mathrm{rand}})}_{\text{total exploitable gap}} = \underbrace{\mathrm{Gap}_{\mathrm{online}}(\mathbf{x}^{\mathrm{alg}})}_{\text{algorithm regret}} + \underbrace{\mathrm{Gap}_{\mathrm{lower}}(\mathbf{x}^{\mathrm{rand}})}_{\text{baseline suboptimality}}$$

The distance from Online to UB is our algorithm's regret. The distance from LB to UB is the total exploitable range. The ratio of these two quantities is the fraction of the available improvement our algorithm captures.

Lower Bound "Random"

A set $\mathcal{PF}^{\mathrm{LB}}$ is a valid Pareto lower bound if and only if:

$$\forall\, \mathbf{x} \in \mathcal{X}_{\mathrm{feasible}}, \quad \exists\, \mathbf{l} \in \mathcal{PF}^{\mathrm{LB}} \text{ such that } \mathbf{l} \trianglelefteq \mathbf{F}(\mathbf{x}) \text{ or } \mathbf{F}(\mathbf{x}) \text{ is non-dominated by } \mathbf{l}$$

- $\forall\, \mathbf{x} \in \mathcal{X}_{\mathrm{feasible}}$: For *every single valid scheduling decision* our system could physically ever make...
- $\exists\, \mathbf{l} \in \mathcal{PF}^{\mathrm{LB}}$: ...there is a point ($\mathbf{l}$) on the Lower Bound...
- $\mathbf{l} \trianglelefteq \mathbf{F}(\mathbf{x})$: ...that is mathematically **worse than or equal to** the results of our decision.

More precisely, for a lower bound in the regret-theoretic sense: we need a mechanism π^{LB} such that for any other mechanism π:

$$\mathcal{PF}(\pi^{\mathrm{LB}}) \trianglelefteq \mathcal{PF}(\pi) \trianglelefteq \mathcal{PF}^{\mathrm{UB}}$$

The random acceptance policy is the correct choice because:

- It makes decisions without future knowledge, just like any online algorithm.
- It is the worst-case policy
- **It is parameter-free**, it requires no tuning, no weights, no thresholds, making it a clean baseline.
- **It is analytically tractable**, its expected objective values have closed-form expressions

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Define the random acceptance policy π^{rand} as:

$$x_j^{\mathrm{rand}} \overset{\mathrm{iid}}{\sim} \mathrm{Bernoulli}(p), \qquad p \in (0,1)$$

Each job is accepted independently with probability p, rejected with probability $1-p$. The decision $x_j^{\mathrm{rand}} \in \{0,1\}$ is made at job arrival time, with no knowledge of future jobs or resource states.

The Fluid Volume constraint for CPU (identical logic applies to RAM):

$$\sum_{j=1}^{N} x_j^{\mathrm{rand}} \cdot A_j^{\mathrm{cpu}} \cdot D_j \leq c_{\mathrm{cpu}} \cdot T$$

Taking expectations under $x_j^{\mathrm{rand}} \sim \mathrm{Bernoulli}(p)$:

$$\mathbb{E}\left[\sum_{j=1}^{N} x_j^{\mathrm{rand}} \cdot A_j^{\mathrm{cpu}} \cdot D_j\right] = p \sum_{j=1}^{N} A_j^{\mathrm{cpu}} \cdot D_j$$

For this to satisfy the constraint in expectation:

$$p \sum_{j=1}^{N} A_j^{\mathrm{cpu}} \cdot D_j \leq c_{\mathrm{cpu}} \cdot T \implies p \leq \frac{c_{\mathrm{cpu}} \cdot T}{\sum_{j=1}^{N} A_j^{\mathrm{cpu}} \cdot D_j}$$

Let $V_i = \sum_j A_j^{(i)} D_j$ be the total volume demand for resource i, and $C_i = c_i \cdot T$ the total volume capacity. Then the maximum feasible probability for resource i is:

$$p_i^{\max} = \frac{C_i}{V_i} = \frac{c_i \cdot T}{\sum_{j=1}^{N} A_j^{(i)} \cdot D_j}$$

The maximum feasible uniform acceptance probability satisfying both constraints simultaneously is:

$$\boxed{p^* = \min\left(\frac{c_{\text{cpu}} \cdot T}{\sum_j A_j^{\text{cpu}} D_j}, \quad \frac{c_{\text{ram}} \cdot T}{\sum_j A_j^{\text{ram}} D_j}, \quad 1 \right)}$$

This is the load factor of the system — the fraction of total demand that capacity can serve if it arrived uniformly.

Intuition: If $p^* = 0.4$, then the system can serve at most 40% of the total arriving volume. A random policy accepting 40% of jobs on average will just fill the cluster.

III.3 — Expected Objective Values Under the Random Policy

Under $x_j \overset{\text{iid}}{\sim} \text{Bernoulli}(p^*)$, the expected objectives are:

$$\mathbb{E}\left[f_1^{\text{sat}}\right] = p^* \sum_{j=1}^{N} q_j v_j = p^* \cdot \mathbf{1}^\top \mathbf{c}_{\text{sat}}$$

$$\mathbb{E}\left[f_2^{\text{prof}}\right] = p^* \sum_{j=1}^{N} (\phi_j - C_j^{\text{elec}}) = p^* \cdot \mathbf{1}^\top \mathbf{c}_{\text{prof}}$$

$$\mathbb{E}\left[f_3^{\text{carb}}\right] = p^* \sum_{j=1}^{N} C_j^{\text{carbon}} = p^* \cdot \mathbf{1}^\top \mathbf{c}_{\text{carb}}$$

But the expected objective vector is a single point — not a Pareto front. To generate the lower-bound Pareto front, we need to run R independent realizations of the random policy and extract the non-dominated envelope.

Generating the Lower-Bound Pareto Front in practice

Monte Carlo Envelope Algorithm
Run R independent realizations of the random policy. For each run r:

1. $\mathbf{x}^{(r)} = (x_1^{(r)}, \ldots, x_N^{(r)}), \quad x_j^{(r)} \overset{\text{iid}}{\sim} \text{Bernoulli}(p^*)$
2. Compute the objective vector:$\mathbf{F}^{(r)} = \left(f_1^{\text{sat}}(\mathbf{x}^{(r)}),\ f_2^{\text{prof}}(\mathbf{x}^{(r)}),\ f_3^{\text{carb}}(\mathbf{x}^{(r)}) \right)$

The empirical lower-bound front is the non-dominated set:

$$\mathcal{PF}^{\text{LB}} = \text{ND}\left(\{\mathbf{F}^{(1)}, \ldots, \mathbf{F}^{(R)}\} \right)$$

How many runs R? The normalized front concentrates around the ray $\mathbf{d}$. By the Central Limit Theorem, the standard deviation of each objective scales as $O(1/\sqrt{N})$ with the number of jobs N. For convergence of the empirical non-dominated envelope, $R \geq 500$ is sufficient for typical cloud workloads with $N \geq 100$ jobs.

A more informative variant: stratified p-sweep

To generate a richer lower-bound front that reveals the shape of the stochastic envelope, sweep p across $[0, p^*]$ in K steps. For each $p_k = k \cdot p^*/K$, run R_k realizations:

$$\mathcal{PF}^{\text{LB}} = \text{ND}\left(\bigcup_{k=0}^{K} \bigcup_{r=1}^{R_k} \{\mathbf{F}^{(k,r)}\} \right)$$

This is the lower-bound Pareto front you report in your thesis.

Computing the Distance between the Three Fronts

The six pairwise distances

$$\begin{array}{ll} d(\mathcal{PF}^{\text{UB}}, \mathcal{PF}^{\text{LB}}) & \text{Total exploitable range} \\ d(\mathcal{PF}^{\text{UB}}, \mathcal{PF}^{\text{Online}}) & \text{Algorithm regret (primary thesis claim)} \\ d(\mathcal{PF}^{\text{Online}}, \mathcal{PF}^{\text{LB}}) & \text{Improvement over baseline} \\ d(\mathcal{PF}^{\text{UB}}, \mathcal{PF}^{\text{UB}}) & = 0 \text{ (sanity check)} \end{array}$$

Before computing any indicator, we need to apply the transformation:

$$\tilde{f}_k^{\min} = 1 - \tilde{f}_k^{\max}, \qquad k \in \{1, 2, 3\}$$

In our normalized space: satisfaction and profit flip to $1 - V_{\text{sat}}$ and $1 - V_{\text{prof}}$; sustainability (already $= -\text{carbon}/z_{\text{carb}}^{\max}$) becomes $1 - V_{\text{sus}}$.

After the transform, all three objectives are minimized, and the Fluid LP front is the best (smallest values) — confirming it is the upper bound.

The Hypervolume Gap (HV-Gap)

In minimization space with reference point $\mathbf{r} = (1.1, 1.1, 1.1)$:

$$\Delta\text{HV}^{\text{regret}} = \text{HV}(\tilde{\mathcal{PF}}^{\text{UB}}, \mathbf{r}) - \text{HV}(\tilde{\mathcal{PF}}^{\text{Online}}, \mathbf{r})$$

$$\Delta\text{HV}^{\text{total}} = \text{HV}(\tilde{\mathcal{PF}}^{\text{UB}}, \mathbf{r}) - \text{HV}(\tilde{\mathcal{PF}}^{\text{LB}}, \mathbf{r})$$

The relative HV-efficiency of the algorithm is:

$$\eta_{\text{HV}} = \frac{\text{HV}(\tilde{\mathcal{PF}}^{\text{Online}}) - \text{HV}(\tilde{\mathcal{PF}}^{\text{LB}})}{\text{HV}(\tilde{\mathcal{PF}}^{\text{UB}}) - \text{HV}(\tilde{\mathcal{PF}}^{\text{LB}})} \in [0, 1]$$

- $\eta_{\text{HV}} = 1$ means the algorithm recovers all the hypervolume the LP captures above the random baseline.
- $\eta_{\text{HV}} = 0$ means the algorithm is no better than random.

IGD⁺ Against the LP Reference Set

$$\text{IGD}^+(\tilde{\mathcal{PF}}^{\text{Online}}, \tilde{\mathcal{Z}}^*) = \frac{1}{|\tilde{\mathcal{Z}}^*|} \sum_{\mathbf{z} \in \tilde{\mathcal{Z}}^*} \min_{\mathbf{a} \in \tilde{\mathcal{PF}}^{\text{Online}}} d^+(\mathbf{z}, \mathbf{a})$$

$$\text{IGD}^+(\tilde{\mathcal{PF}}^{\text{LB}}, \tilde{\mathcal{Z}}^*) = \frac{1}{|\tilde{\mathcal{Z}}^*|} \sum_{\mathbf{z} \in \tilde{\mathcal{Z}}^*} \min_{\mathbf{a} \in \tilde{\mathcal{PF}}^{\text{LB}}} d^+(\mathbf{z}, \mathbf{a})$$

The IGD⁺ improvement ratio of the algorithm over the random baseline:

$$\rho_{\text{IGD}^+} = 1 - \frac{\text{IGD}^+(\tilde{\mathcal{PF}}^{\text{Online}}, \tilde{\mathcal{Z}}^*)}{\text{IGD}^+(\tilde{\mathcal{PF}}^{\text{LB}}, \tilde{\mathcal{Z}}^*)}$$

- $\rho = 1$ means the online algorithm achieves zero distance to the LP front (impossible).
- $\rho = 0$ means no improvement over random.

The ε-Indicator Worst-Case Gap

$$I_{\varepsilon^+}(\tilde{\mathcal{PF}}^{\text{Online}}, \tilde{\mathcal{Z}}^*) = \max_{\mathbf{z} \in \tilde{\mathcal{Z}}^*} \min_{\mathbf{a} \in \tilde{\mathcal{PF}}^{\text{Online}}} \max_k (a_k - z_k)$$

This is the worst-case objective-wise shortfall — the hardest LP point the online mechanism fails to dominate, and by how much. Analogously for the random front.

Metric	PFLB vs PFUB	PFOnline vs PFUB	Improvement ρ
HV $\uparrow$	h_{LB}	h_{OL}	$\eta_{\text{HV}} = \frac{h_{\text{OL}} - h_{\text{LB}}}{h_{\text{UB}} - h_{\text{LB}}}$
IGD⁺ $\downarrow$	δ_{LB}	δ_{OL}	$\rho = 1 - \delta_{\text{OL}}/\delta_{\text{LB}}$
I_{ε^+} $\downarrow$	ε_{LB}	ε_{OL}	$1 - \varepsilon_{\text{OL}}/\varepsilon_{\text{LB}}$